

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

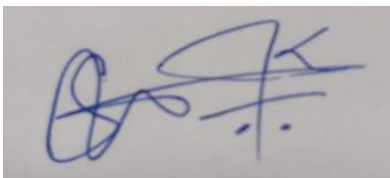
Problem Statement: INTELLIGENT E-GOVERNANCE CITIZEN SERVICE REQUEST MANAGEMENT SYSTEM

Course Code	CSA10
Course Title	Software Engineering
CO Assessed	CO4
Assessment	Assessment Tool 1 – Test Case Design Assignment
Weightage	20%
Total Marks	25
CO4 Statement	Implement robust testing, quality assurance, and DevOps practices, emphasising security, reliability, and ethics
Student Name	S SUVIKUMAR
Reg. No.	192424262
Date	
Duration	60 minutes

Code of Conduct

I S Suvikumar (192424262) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification (15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness (25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results (20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques (15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage (15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation (10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Marks Summary

Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

Annexure A – Test Case Design Document

INTELLIGENT E-GOVERNANCE CITIZEN SERVICE REQUEST MANAGEMENT SYSTEM

1. Problem Statement Overview

INTELLIGENT E-GOVERNANCE CITIZEN SERVICE REQUEST MANAGEMENT SYSTEM is an IoT-enabled software platform that helps fish farmers and aquaculture administrators monitor pond water quality (pH, temperature, dissolved oxygen, ammonia), manage feeding schedules, track fish growth and health records, and receive real-time alerts when water conditions become unsafe. The system supports farmer and admin roles, multi-pond management, report generation, and mobile/SMS notifications for critical events.

2. Functional Requirements (FR)

- FR1: The system shall allow farmers/admins to register and log in securely using credentials.
- FR2: The system shall acquire real-time water quality data (pH, temperature, dissolved oxygen, ammonia) from IoT sensors installed in each pond.
- FR3: The system shall display real-time and historical sensor readings on a dashboard.
- FR4: The system shall automatically generate alerts/notifications when water quality parameters exceed safe thresholds.
- FR5: The system shall allow farmers to create, edit, and view fish feeding schedules.
- FR6: The system shall allow farmers to record and track fish growth and health data over time.
- FR7: The system shall allow an admin to manage multiple ponds/farms under a single account.
- FR8: The system shall generate exportable reports (PDF/Excel) for water quality trends, feeding history, and growth records.
- FR9: The system shall allow users to reset a forgotten password using a registered email/OTP.
- FR10: The system shall send SMS/app push notifications to the farmer's registered mobile number for critical alerts.

3. Non-Functional Requirements (NFR)

- NFR1 (Performance): The dashboard shall reflect updated sensor data within 5 seconds of data receipt.
- NFR2 (Reliability): The system shall maintain at least 99.5% uptime for continuous pond monitoring.
- NFR3 (Usability): The dashboard shall be intuitive for farmers with minimal technical literacy and support multilingual display.
- NFR4 (Security): All user credentials and sensitive data shall be encrypted (AES-256) and access shall be role-based.
- NFR5 (Scalability): The system shall support simultaneous monitoring of up to 500 ponds without performance degradation.
- NFR6 (Availability of Alerts): Critical alert notifications shall be delivered within 30 seconds of a threshold breach.

4. Test Scenarios

- TS1: Verify user registration, login, and authentication.
- TS2: Verify acquisition and display of real-time sensor data.
- TS3: Verify alert generation for abnormal water quality parameters (pH, temperature, DO).
- TS4: Verify feeding schedule creation, validation, and reminders.
- TS5: Verify fish growth/health record entry and retrieval.
- TS6: Verify multi-pond management functionality for admin users.
- TS7: Verify report generation and export (PDF/Excel).
- TS8: Verify password reset workflow via OTP.
- TS9: Verify SMS/app critical alert notification delivery.
- TS10: Verify system behavior during sensor failure or connectivity loss.

5. Application of Test Design Techniques

Four complementary test design techniques were applied to derive high-value test cases with minimum redundancy, shown below.

5.1 Equivalence Partitioning (EP) — applied to sensor and input parameters:

Parameter	Invalid Partition (Low)	Valid Partition	Invalid Partition (High)
Pond pH	pH < 6.5	6.5 – 8.5	pH > 8.5
Water Temperature (°C)	Temp < 24°C	24°C – 30°C	Temp > 30°C
Dissolved Oxygen (mg/L)	DO < 4 mg/L	4 – 8 mg/L	DO > 8 mg/L

Parameter	Invalid Partition (Low)	Valid Partition	Invalid Partition (High)
Password length	< 8 characters	8 – 16 characters	> 16 characters
Feed quantity (kg)	≤ 0 kg	0.1 – 50 kg	> 50 kg

5.2 Boundary Value Analysis (BVA) — applied to threshold limits:

Parameter	Lower Boundary – 1	Lower Boundary	Upper Boundary	Upper Boundary + 1
Pond pH	6.4 (Alert)	6.5 (Safe)	8.5 (Safe)	8.6 (Alert)
Water Temperature	23°C (Alert)	24°C (Safe)	30°C (Safe)	31°C (Alert)
Dissolved Oxygen	3.9 mg/L (Alert)	4.0 mg/L (Safe)	8.0 mg/L (Safe)	8.1 mg/L (Alert)
Feed quantity	-0.1 kg (Rejected)	0.1 kg (Accepted)	50 kg (Accepted)	50.1 kg (Rejected)
Pond count (Admin)	-	1 pond (Accepted)	500 ponds (Accepted)	501 ponds (Rejected)

5.3 Decision Table Testing — applied to the alert-generation logic (combination of pH, temperature, and DO status):

Rule	pH Normal?	Temperature Normal?	DO Normal?	System Action
R1	Y	Y	Y	No alert – status Normal
R2	N	Y	Y	Warning alert – pH out of range
R3	Y	N	Y	Warning alert – temperature out of range
R4	Y	Y	N	Warning alert – low dissolved oxygen
R5	N	N	Y	Critical alert + dashboard flag
R6	N	N	N	Critical alert + SMS/app notification sent

5.4 State Transition Testing — applied to the pond monitoring/alert lifecycle:

Current State	Trigger Event	Next State
Normal	One parameter breaches threshold	Warning
Warning	Additional parameter(s) breach threshold	Critical
Warning	Parameter returns to safe range	Normal
Critical	Alert dispatched to farmer (SMS/app)	Alert Sent
Alert Sent	Farmer acknowledges / parameters normalize	Resolved
Resolved	New reading received	Normal

6. Detailed Test Cases

The following 30 test cases cover normal, boundary, invalid, and exceptional conditions across all major modules of the system.

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC01	Valid user login	1. Open app 2. Enter valid username/password 3. Tap Login	user: farmer01 / pwd: Fish@2026	User logged in; dashboard displayed	As expected	Pass
TC02	Invalid login password	1. Open app 2. Enter valid username, wrong password 3. Tap Login	user: farmer01 / pwd: wrong123	Error message 'Invalid credentials' displayed	As expected	Pass
TC03	Password length lower boundary (valid)	1. Register 2. Enter password of exactly 8 characters	pwd: Fish@123	Password accepted	As expected	Pass
TC04	Password length below boundary (invalid)	1. Register 2. Enter password of 7 characters	pwd: Fis@12	Error: 'Password must be at least 8 characters'	As expected	Pass

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC05	Login with no network connectivity	1. Disable network 2. Attempt login	user: farmer01 / pwd: Fish@2026	Error: 'No internet connection, please retry' shown; no crash	As expected	Pass
TC06	Sensor reading within safe pH range	1. Sensor sends pH reading 2. View dashboard	pH = 7.2	Dashboard shows 'Normal' status (green), no alert raised	As expected	Pass
TC07	pH at lower safe boundary	1. Sensor sends pH = 6.5 2. View dashboard	pH = 6.5	Status remains 'Normal'; no alert	As expected	Pass
TC08	pH just below lower boundary	1. Sensor sends pH = 6.4 2. View dashboard	pH = 6.4	Status changes to 'Warning'; alert generated	As expected	Pass
TC09	pH at upper safe boundary	1. Sensor sends pH = 8.5 2. View dashboard	pH = 8.5	Status remains 'Normal'; no alert	As expected	Pass
TC10	pH just above upper boundary	1. Sensor sends pH = 8.6 2. View dashboard	pH = 8.6	Status changes to 'Warning'; alert generated	As expected	Pass
TC11	Sensor sends invalid/out-of-range value	1. Sensor transmits corrupt reading 2. View dashboard	pH = -5 (invalid)	System flags 'Sensor Error'; value discarded; app does not crash	As expected	Pass
TC12	Sensor connectivity lost	1. Disconnect sensor for > 5 minutes 2. View dashboard	No data received for 5+ min	Dashboard shows 'No Data / Sensor Offline'; connectivity alert sent	As expected	Pass
TC13	All parameters normal (decision R1)	1. Sensors send readings 2. View dashboard status	pH=7.0, Temp=27°C, DO=6 mg/L	Status 'Normal'; no alert raised	As expected	Pass
TC14	Single parameter abnormal (decision R2)	1. Sensors send readings 2. View dashboard status	pH=6.0, Temp=27°C, DO=6 mg/L	Warning alert for pH only	As expected	Pass
TC15	Two parameters abnormal (decision R5)	1. Sensors send readings 2. View dashboard status	pH=6.0, Temp=33°C, DO=6 mg/L	Critical alert with pH and temperature flagged	As expected	Pass
TC16	All parameters abnormal (decision R6)	1. Sensors send readings 2. Check notifications	pH=6.0, Temp=33°C, DO=3 mg/L	Critical alert raised; SMS/app notification sent to farmer within 30 sec	As expected	Pass
TC17	Create valid feeding schedule	1. Open feeding module 2. Enter time & quantity 3. Save	Time: 7:00 AM, Qty: 10 kg	Schedule saved; reminder set for 7:00 AM	As expected	Pass
TC18	Enter negative feed quantity	1. Open feeding module 2. Enter negative quantity 3. Save	Qty: -5 kg	Validation error: 'Quantity must be greater than 0'	As expected	Pass
TC19	Feed quantity at maximum boundary	1. Enter feed quantity = 50 kg 2. Save	Qty: 50 kg	Accepted; schedule saved	As expected	Pass
TC20	Feed quantity exceeds maximum	1. Enter feed quantity = 51 kg 2. Save	Qty: 51 kg	Rejected; error 'Quantity exceeds permissible limit (50 kg)'	As expected	Pass
TC21	Add valid fish growth record	1. Open growth module 2. Enter weight & date 3. Save	Weight: 250 g, Date: 15-Aug-2026	Record saved; growth chart updated	As expected	Pass
TC22	Enter future date for growth record	1. Open growth module 2. Enter a future date 3. Save	Date: 30-Dec-2027	Validation error: 'Date cannot be in the future'	As expected	Pass
TC23	Generate water quality report	1. Select date range 2. Tap 'Generate Report'	Range: 01-Aug-2026 to 15-Aug-2026	PDF/Excel report generated with correct data for range	As expected	Pass
TC24	Generate report with no data in range	1. Select a date range with no records 2. Generate	Range: 01-Jan-2020 to 05-Jan-2020	Message: 'No data found for selected range'	As expected	Pass

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
TC25	Password reset – valid OTP request	1. Tap 'Forgot Password' 2. Enter registered email 3. Request OTP	email: farmer01@example.com	OTP sent to registered email within 30 seconds	As expected	Pass
TC26	Password reset – incorrect OTP	1. Enter incorrect OTP 2. Submit	OTP entered: 000000 (wrong)	Error: 'Invalid OTP'; attempt counter incremented	As expected	Pass
TC27	Account lock after repeated OTP failures (state transition)	1. Enter wrong OTP 3 times consecutively	3 invalid OTP attempts	Account state changes from 'Active' to 'Locked'; further attempts blocked for 15 minutes	As expected	Pass
TC28	Admin adds a new pond	1. Login as admin 2. Tap 'Add Pond' 3. Enter pond details 4. Save	Pond name: Pond-12, Area: 2 acres	Pond added successfully; visible on dashboard	As expected	Pass
TC29	Admin adds pond at maximum boundary	1. Add the 500th pond to account	Pond count = 500	Accepted; 500th pond added successfully	As expected	Pass
TC30	Admin exceeds maximum pond limit	1. Attempt to add a 501st pond	Pond count = 501	Rejected; error 'Maximum pond limit (500) reached'	As expected	Pass

7. Traceability Matrix

The matrix below maps each functional/non-functional requirement to its corresponding test scenario and detailed test case IDs, ensuring complete requirement coverage.

Requirement ID	Requirement Summary	Test Scenario	Related Test Case IDs
FR1	Secure registration & login	TS1	TC01, TC02, TC03, TC04, TC05
FR2 / NFR1	Real-time sensor data acquisition	TS2	TC06, TC07, TC09, TC11
FR4 / NFR6	Automatic alert generation	TS3	TC08, TC10, TC13, TC14, TC15, TC16
FR5	Feeding schedule management	TS4	TC17, TC18, TC19, TC20
FR6	Fish growth/health tracking	TS5	TC21, TC22
FR7 / NFR5	Multi-pond management (Admin)	TS6	TC28, TC29, TC30
FR8	Report generation & export	TS7	TC23, TC24
FR9 / NFR4	Password reset via OTP	TS8	TC25, TC26, TC27
FR10 / NFR6	SMS/app critical notifications	TS9	TC16
NFR2	System reliability / connectivity handling	TS10	TC12

8. Test Coverage Summary

- Total Functional Requirements identified: 10 (FR1–FR10) — all mapped to at least one test case.
- Total Non-Functional Requirements identified: 6 (NFR1–NFR6) — all addressed through performance, reliability, security, and availability-oriented test cases.
- Total Test Scenarios: 10 (TS1–TS10).
- Total Test Cases designed: 30, covering normal, boundary, invalid, and exceptional conditions.
- Test design techniques applied: Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing.
- Overall requirement traceability achieved: 100% (every FR/NFR maps to a test scenario and test case).